Matrix Theory: Lecture 3

Carl Olsson

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From Linear Algebra

$e_1, e_2, ..., e_n$ and $\widehat{e}_1, \widehat{e}_2, \ldots, \widehat{e}_n$ two bases of $\mathbb{R}^n$

$$\widehat{e}_j = \sum_i s_{ij} e_i.$$

X - coordinates in $e_1, \ldots, e_n$
$\hat{X}$ - coordinates in $\widehat{e}_1, \ldots, \widehat{e}_n$
S transition matrix

$$X = S\hat{X} \Leftrightarrow \hat{X} = S^{-1}X.$$

A matrix representation of lin. op. $\mathcal{A}$ in basis $e_1, e_2, .., e_n$
$\hat{A}$ matrix representation of lin. op. $\mathcal{A}$ in basis $e_1, e_2, .., e_n$

$$\hat{A} = S^{-1}AS$$

Def. Two matrices A and B are called **similar** if $B = S^{-1}AS$ for some invertible matrix S.

Ex. of questions that we will study in Chapters 6 and 7

When is a matrix A diagonalizable? $\Leftrightarrow$
When is a A similar to some diagonal D? $\Leftrightarrow$
When does $\mathcal{A}$ have a diagonal matrix representation D?
If a matrix is not diagonalizable is there any generalization?
(Jordanization, ch 7)

Need tools that are invariant to choice of basis.
(Developed in Chapters 4 and 5.)

Outline

- General Vector Spaces
- Invariant concepts: independence, bases, dimension
- Linear maps between vector spaces
- Similarity, image, kernel
- Invariant concepts: rank, determinant, trace

Field

A **field** is a set $\mathbb{K}$ with addition, $+$, and multiplication between elements (the results being in $\mathbb{K}$ too) with the following properties:

- *commutativity*: $a + b = b + a$; $ab = ba$,
- *associativity*: $a + (b + c) = (a + b) + c$; $(ab)c = a(bc)$,
- *distributivity*: $a(b + c) = ab + ac$,
- *existence of zero and unity*: there are two elements $0 \neq 1 \in \mathbb{K}$ such that $a + 0 = a$, $a \cdot 1 = a$ for any $a \in \mathbb{K}$,
- *solvability of equations*: every linear equation $ax + b = c$ has a unique solution for $a \neq 0$.

Subtraction is obtained through $c - b$ being the only solution of the equation $x + b = c$.
Division b/a for $a \neq 0$ as the only solution of the equation $ax = b$.

Vector Space

Let $\mathbb{K}$ be a field. A **vector space** over $\mathbb{K}$ is a set V with addition, $+$, between elements in V called **vectors** and multiplications between vectors and numbers in $\mathbb{K}$ and the following properties:

- commutativity: $v + u = u + v$,
- associativity: $v + (u + w) = (v + u) + w$; $(c_1 c_2)v = c_1(c_2 v)$,
- distributivity: $c(v + u) = cv + cu$; $(c_1 + c_2)v = c_1 v + c_2 v$,
- existence of zero vector: there exists a vector $0 \in V$ such that $v + 0 = v$ for any $v \in V$,
- solvability of equations: every equation $x + v = u$ has a unique solution which is denoted $u - v$.
- $1v = v, 0v = 0$ for any $v \in V$.

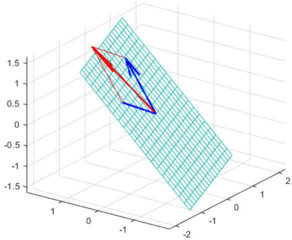
Here $c, c_1, c_2 \in \mathbb{K}$; $v, u, w \in V$.

Subspaces

Let V be a vector space over $\mathbb{K}$. The subset $U \subseteq V$ is called a subspace if for any $u, v \in U$ and $c \in \mathbb{K}$ we have

$$u + v \in U, cv \in U.$$

We automatically get that a subspace is a vector space by itself with the same operations as in V.



Linear combination, dependence, basis

- If $v_1, v_2, ..., v_n \in V$ and $x_1, x_2, \ldots, x_n$ the expression

$$x_1 v_1 + x_2 v_2 + \ldots + x_n v_n$$

  is called a **linear combination**.

- If any $v \in V$ can be written

$$v = x_1 v_1 + x_2 v_2 + \ldots + x_n v_n$$

  for some $x_1, x_2, ..., x_n \in \mathbb{K}$ then $v_1, v_2, ..., v_n$ **spans/generates** V and V finite dimensional. If $x_1, x_2, \ldots, x_n$ are unique for each $v \in V$ then $v_1, v_2, ..., v_n$ forms a **basis** of V and $x_1, x_2, ..., x_n$ are the **coordinates** of v in that basis.

- A set of vectors $v_1, v_2, \ldots, v_n$ are called **linearly independent** if

$$x_1 v_1 + x_2 v_2 + \ldots + x_n v_n = 0 \Rightarrow x_1 = x_2 = \ldots = x_n = 0.$$

We will only consider finite dimensional spaces in the course.

Linear combination, dependence, basis

Testing linear independence with coordinates: If $e_1, \ldots, e_m$ is a basis and $v_i = \sum_j a_{ji} e_j$ (A_i contains coord. of v_i) then

$$\sum_i x_i v_i = 0 \Leftrightarrow \sum_i x_i \sum_j a_{ji} e_j = 0 \Leftrightarrow \sum_j \underbrace{\left(\sum_i a_{ji} x_i \right)}_{=0} e_j = 0.$$

Therefore

$$\begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1k} \\ a_{21} & a_{22} & \ldots & a_{2k} \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mk} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \ldots \\ x_k \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \ldots \\ 0 \end{pmatrix} \Leftrightarrow$$

$$x_1 A_1 + \ldots + x_k A_k = 0.$$

Dimension of a Vector Space

Theorem 4.4

All bases in a finite dimensional vector space V have the same number of vectors. This number n is called its **dimension**, $\dim V$.

Lemma 4.7

If the matrix A has a left inverse then the column vectors $X_1, \ldots, X_k$ are linearly independent if and only if the vectors $AX_1, \ldots, AX_k$ are linearly independent.

Theorem 4.8

Every set of linearly independent vectors can be completed to a basis.

Corollary 4.9

If U is a subspace in V then $\dim U \leq \dim V$.
If $\dim U = \dim V$ then $U = V$.

Linear Maps

Def. If U, V are vectorspaces over the same field $\mathbb{K}$ a map $F : V \longrightarrow U$ is called **linear** if

$$F(v+v') = F(v) + F(v'), F(cv) = cF(v)$$

for any $v, v' \in V$ and $c \in \mathbb{K}$.
In the case $U = V$ one usually says operator instead of map.

Any linear map $F : V \rightarrow U$ where U and V are finite can be represented by a matrix A. Let $e_1, e_2, \ldots, e_n$ be a basis of V and $f_1, f_2, \ldots, f_m$ of U. The matrix $A = \begin{pmatrix} A_1 & A_2 & \ldots & A_n \end{pmatrix}$, where column A_i contains the coordinates of $F(e_i)$ in the basis $f_1, f_2, \ldots, f_m$ describes the mapping.

Similar Matrices and Linear Operators

If we characterize an operator $\mathcal{A}$ through properties of its matrix representation $S^{-1}AS$ we need to make sure that these properties do not change with S.

Theorem 5.4

If A and B are similar then

- A^T and B^T are similar,
- cA and cB are similar,
- A^k and B^k are similar,
- $p(A)$ and $p(B)$ are similar for any polynomial,
- $|A| = |B|$,
- $\text{tr}(A) = \text{tr}(B)$,
- $r(A) = r(B)$.

Thus suggests that rank, tr etc. can be defined for an operator directly.

Image of a Linear Map

Def. If $\mathcal{A} : V \to U$ and $W \subseteq V$ we let $\mathcal{A}W = \{\mathcal{A}w | w \in W\}$.

Theorem 5.6

If $W \subseteq V$ is a subspace then $\mathcal{A}W \subseteq U$ is also a subspace and

$$\dim \mathcal{A}W \leq \dim W.$$

If $\mathcal{A}$ has a left inverse then

$$\dim \mathcal{A}W = \dim W.$$

Image of a Linear Map

Def. The subspace $\mathcal{A}V$ is called the **image** of the linear map $\mathcal{A} : V \to U$ and is denoted $\operatorname{Im} \mathcal{A}$. In other words,

$$\operatorname{Im} \mathcal{A} = \{\mathcal{A}x | x \in V\}$$

is the set of all vectors that may be obtained applying $\mathcal{A}$.
We have the same notation for the **image of the matrix**:

$$\operatorname{Im} A = \{AX | X \in \mathbb{K}^n\},$$

where $n = \dim V$.

Rank of a Linear Map

Def. For a linear map $\mathcal{A}$ we let

$$\operatorname{rank} \mathcal{A} = \dim(\operatorname{Im} \mathcal{A})$$

(Invariant definition: depends only on the operator not the matrix representation.)

For a matrix A we now have two definitions:

1. The number of pivot elements of U if $PA = LU$.
2. $\operatorname{rank}(\mathcal{A})$ where $\mathcal{A}$ is the linear map $X \mapsto AX$.

Do they coincide?

The Kernel of a Linear Map

Def. The subspace

$$\operatorname{Ker}\mathcal{A} = \{v \in V | \mathcal{A}v = 0\}$$

is called the **kernel** of the linear map $\mathcal{A}$.
For a matrix A its kernel is therefore the set of all solutions X of the system of equations $AX = O$.

The Kernel of a Linear Map

Theorem 5.12

Let $\mathcal{A} : V \to U$ be a linear map. Then

$$\dim \operatorname{Im} \mathcal{A} + \dim \operatorname{Ker} \mathcal{A} = \dim V.$$

If A is a $m \times n$ matrix then

$$r(A) + \dim \operatorname{Ker} A = n.$$

(Note that n is equal to the number of columns. The number of rows does not matter.)